DRAM Row-Buffer Management Policies

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Row-buffer hit/miss/conflict

Row-buffer	When	Latency
Hit	The row we are requesting is open	ReadWrite
Miss	No row is currently open	Activate+ReadWrite
Conflict	The row we are requesting is NOT open, but some other row is open.	Precharge+Activate+ReadWrite

Precharge = closing row
Activate = opening row (same as RAS)
ReadWrite = reading or writing data (same as CAS)

Row Buffer Management Policies

- Open row
  - Keep the row open after an access

  + Next access might need the same row → row-buffer hit

  -- Next access might need a different row → row-buffer conflict, extra latency

- Closed row
  - Close the row after an access **blindly. That means, without seeing which requests are currently in the queue, we will close the row. Even if next request is to the same row, we will still close the row and open again.**
  - **We are making this assumption in this course for simplicity.**

  + Next access might need a different row → avoid a row-buffer conflict

  -- Next access might need the same row → extra activate latency

Solved Question

- DRAM latencies:
- Opening a row: 30 ns
- Closing a row: 30ns
- Reading/writing data: 30ns
- Initially, no row is open.
- For requests below, find the time when the request is served. That is, the time when the data requested by that request has been provided.
- Assume column-number is 0 for all cases.

- Hint: Open-row policy has only row-buffer hit or row-buffer conflict. [On the first access, if the row buffer is empty, then a row-buffer miss can also happen]
- Hint: Close-row policy has only row-buffer miss.

Row-id	Request arrival time	Open	Close
40	0		
41	30		
40	150		
40	290		
41	310		
40	450		

Open Row policy

- No row is open. There is access to row 40
- This is a first access, so we open row 40
- We incur missLat 60 The access completes at 60

- row 40 is open. There is access to row 41
- It is a row-buffer conflict. TimeOfArrival 30 is less than the time when previous access completed, which is 60. Add conflictLat of 90 to completion time 60. This access will complete on 150

- row 41 is open. There is access to row 40
- It is a row-buffer conflict. TimeOfArrival 150 is equal or more than the time when previous access completed, which is 150. Add conflictLat of 90 to TimeOfArrival 150. This access will complete on 240

Row-id	Request arrival time	Open	Close
40	0	60	
41	30	150	
40	150	240	
40	290		
41	310		
40	450		

Open Row policy

- row 40 is open. There is access to row 40
- This is row-buffer Hit. TimeOfArrival 290 is equal or more than the time when previous access completed, which is 240. Add HitLat of 30 to TimeOfArrival 290. This access will complete on 320

- row 40 is open. There is access to row 41
- It is a row-buffer conflict. TimeOfArrival 310 is less than the time when previous access completed, which is 320. Add conflictLat of 90 to completion time 320. This access will complete on 410

- row 41 is open. There is access to row 40
- It is a row-buffer conflict. TimeOfArrival 450 is equal or more than the time when previous access completed, which is 410. Add conflictLat of 90 to TimeOfArrival 450. This access will complete on 540

Row-id	Request arrival time	Open	Close
40	0	60	
41	30	150	
40	150	240	
40	290	320	
41	310	410	
40	450	540	

Closed row policy

- No row is open.At time 0, there is access to row 40
- This is a first access, so we open row 40
- We incur missLat 60 The access completes at 60. Row will be closed at 90

- No row is open.At time 30, there is access to row 41
- We begin servicing this request from time 90 The request is served at time 150. Row will be closed at 180

- No row is open.At time 150, there is access to row 40
- We begin servicing this request from time 180 The request is served at time 240. Row will be closed at 270

Row-id	Request arrival time	Open	Close
40	0	60	60
41	30	150	150
40	150	240	240
40	290	320	
41	310	410	
40	450	540	

Closed row policy

- No row is open.At time 290, there is access to row 40
- We begin servicing this request from time 290. The request is served at time 350. Row will be closed at 380

- No row is open.At time 310, there is access to row 41
- We begin servicing this request from time 380. The request is served at time 440. Row will be closed at 470

- No row is open.At time 450, there is access to row 40
- We begin servicing this request from time 470. The request is served at time 530. Row will be closed at 560

Row-id	Request arrival time	Open	Close
40	0	60	60
41	30	150	150
40	150	240	240
40	290	320	350
41	310	410	440
40	450	540	530